What Is meant by an autoregressive model?

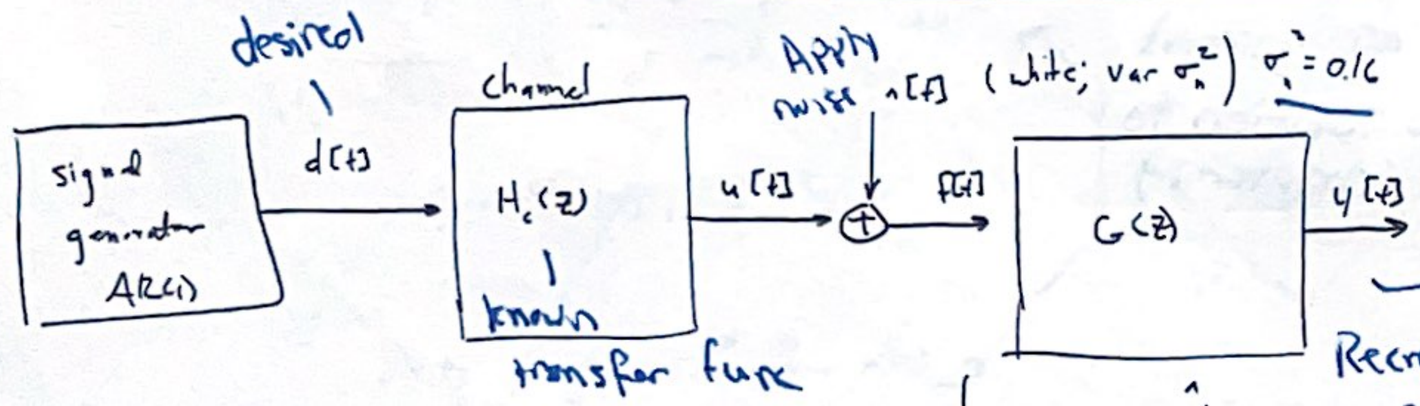
↳

$$x[t-m] = -\left(\sum_{i=1}^{m} b_i\, x[t-m+i]\right) + b[t]$$

$$= -\left(b_1\, x[t-m+1] + b_2\, x[t-m+2] + \dots b_m\, x[t]\right) + b[t]$$

Equalization Problem

Take a signal, pass it through a filter (channel), add noise, then get back original signal. How do we do that?

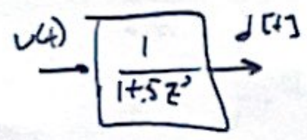


Going to make G(z) a 2-tap equilizer filter

Recreate the signal

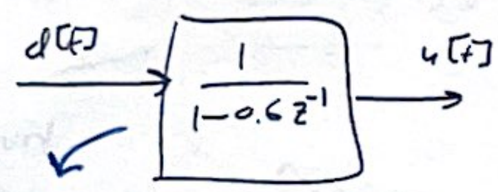
Assume: $d[t]$ is an AR(1) signal.

$$d[t] = -.5\, d[t-1] + v[t]$$



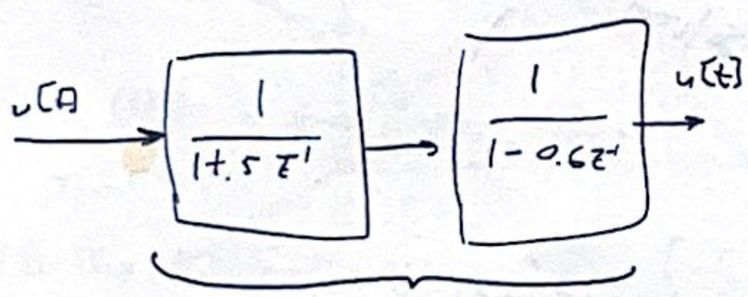
$\sigma_v^2 = 0.1$

$$H_c(z) = \frac{1}{1-0.6z^{-1}}$$



$$u[t] = 0.6\, u[t-1] + d[t]$$

We can think of the signal generator as the combination of white noise $v[t]$ as the input to AR(1)



AR(1) signal generator

$$H(z) = \left(\frac{1}{1+.5z^{-1}}\right)\left(\frac{1}{1-.6z^{-1}}\right) = \frac{1}{1-0.1z^{-1}-.3z^{-2}}$$

rewrite

$$= \frac{1}{1+a_1 z + a_2 z^{-2}}$$

$$y[t] = \hat{d}[t]$$

$$= h_0 f[t] + h_1 f[t-1]$$

choose $\{h_0, h_1\}$ to minimize $E[|d[t] - \hat{d}[t]|^2]$.

Apply orthogonal principle

① $E[(d[t] - (h_0 f[t] + h_1 f[t-1]))\, f[t]] = 0$

② $E[(d[t] - (h_0 f[t] + h_1 f[t-1]))\, f[t-1]] = 0$

$$\Downarrow$$

$$\underbrace{\begin{bmatrix} r_{ff}(0) & r_{ff}(1) \\ r_{ff}(1) & r_{ff}(0) \end{bmatrix}}_{R_{ff}} \begin{bmatrix} h_0 \\ h_1 \end{bmatrix} = \underbrace{\begin{bmatrix} E[d[t]\, f[t]] \\ E[d[t]\, f[t-1]] \end{bmatrix}}_{P}$$

$$f[t] = u[t] + n[t]$$

$$R_{ff} = R_{uu} + R_{nn}$$

$R_{uu} + R_{nn}$

$$R_{uu} = \begin{bmatrix} r_{uu}(0) & r_{uu}(1) \\ r_{uu}(1) & r_{uu}(0) \end{bmatrix}$$

$$R_{nn} = \sigma_n^2 I$$

$r_{uu}(0) = E[u(n)u(n)]$ where $u(n)$ is a AR(2) random process. ~~If u(n) is an~~

where $u(n)$ is an AR(2) r.p

$v_k \perp$

$r_{uu}(0) = E[u(n)u(n)]$ where $u(n)$ is an $AR(2)$ r.p.

$r_{uu}(1) = E[u(n)u(n-1)]$ "

In an earlier assignment we found the solution to this problem (chapter 1)

$$r_{uu}(0) = \left(\frac{1+a_2}{1-a_2}\right)\left(\frac{\sigma_v^2}{(1+a_2)^2 - a_1^2}\right) = 0.1122 \qquad (1)$$

$$r_{uu}(1) = \frac{-a_1}{1+a_2}\sigma_v^2 = 0.016 \qquad (2)$$

Now that we know R_{ff}, we need to find P (see top right page 1)

$$R_{ff} = \underbrace{\begin{bmatrix} .1122 & .016 \\ .016 & .1122 \end{bmatrix}}_{R_{uu}} + 0.16\begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} = \begin{bmatrix} .2722 & .016 \\ .016 & .2722 \end{bmatrix}$$

$E[f(t)\,d(t)] = E[(u(t)+n(t))\,d(t)]$

$= E[u(t)\,d(t)]$

$E[f(t-1)\,d(t)]$

$= E[u(t-1)\,d(t)]$

How do we get these?

Channel equation solved for d

$$u(t) - 0.6\,u(t-1) = d(t)$$

① Mult. by $u(t-k)$:
② $E[\quad]$

$$E[u(t)u(t-k)] - E[0.6\,u(t-1)u(t-k)] = E[u(t-k)\,d(t)]$$

At $k=0$: $\underbrace{E[u(t)^2]}_{r_{uu}(0)} - E[0.6\,u(t-1)u(t)] = \underbrace{E[u(t)\,d(t)]}_{E[u(t)\,d(t)] = .1026}$

$r_{uu}(0) - 0.6\,r_{uu}(1) = E[u(t)\,d(t)]$ Look @ (1)

At $k=1$: $E[u(t)u(t-1)] - 0.6\,E[u(t-1)^2] = \underbrace{E[u(t-1)\,d(t)]}_{r_{uu}(1) - 0.6\,r_{uu}(0) = -.0513}$

Look @ (2)

$$P = \begin{bmatrix} 0.1026 \\ -0.0513 \end{bmatrix}$$

$\underline{h} = R_{ff}^{-1} P$

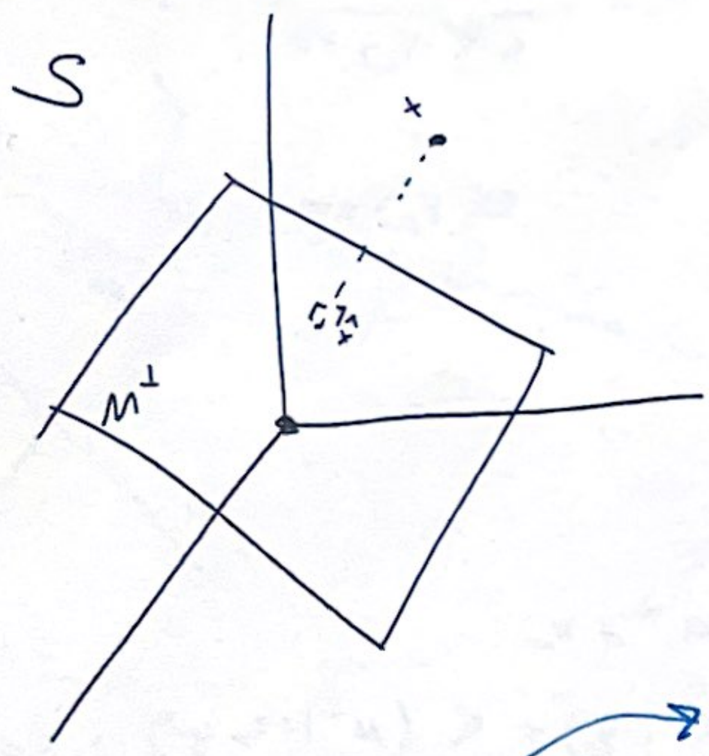
Linear variety

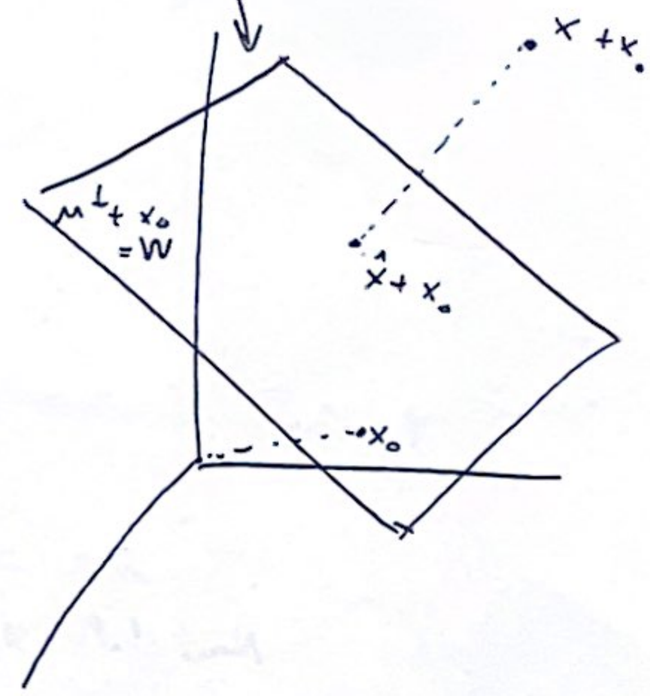
$\underline{y} = K_{FF} F$

Linear variety

No longer a vector space because it does not include the origin (in general)

Dual Approximation





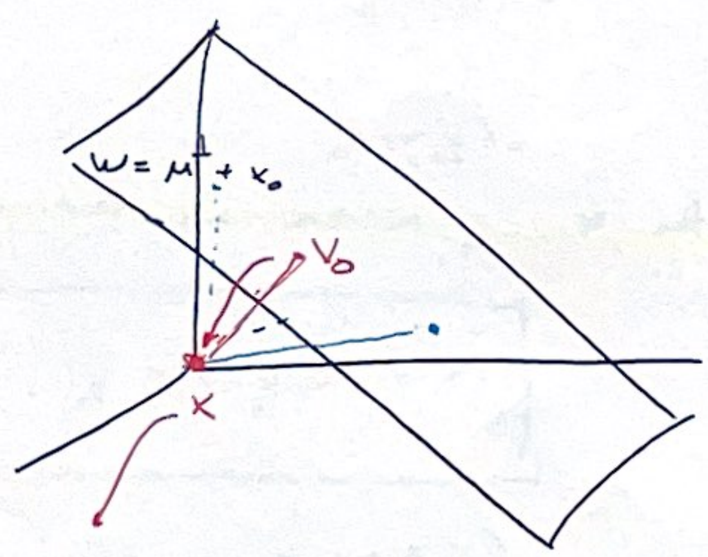
Move plane by some vector x_0

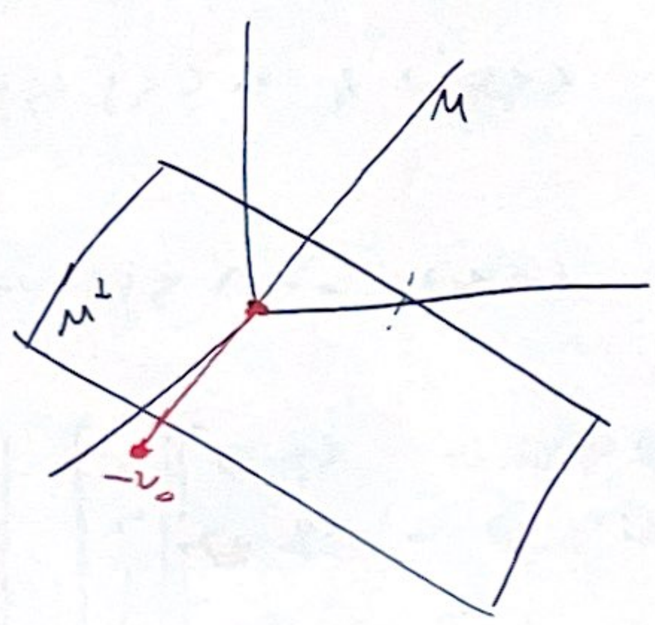
Def: Let $M^\perp$ be a subspace of a v.s. S.

Let $x_0 \in S$.

Form $W = x_0 + M^\perp$

W is a linear variety.





There is a unique vector v_0 in W of min. norm.

and $v_0 \perp M^\perp$

so $v_0 \in M$

$\underline{M} = \text{span}(y_1, y_2, \ldots, y_m)$

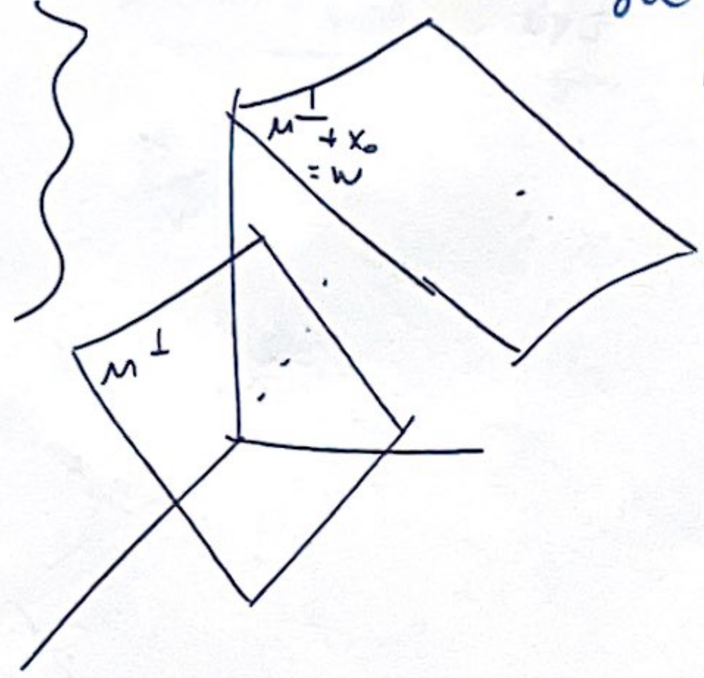
$\underline{M} = \text{span}(y_1, y_2, \ldots, y_m)$

M is $\perp$ to $M^\perp$,

see prev page for M

Let $x \in M^\perp$

$$\left.\begin{aligned} \langle x, y_1\rangle &= 0 \\ \langle x, y_2\rangle &= 0 \\ &\vdots \\ \langle x, y_m\rangle &= 0 \end{aligned}\right\}$$



Now let $x \in M^\perp + x_0$

$$\left.\begin{aligned} \langle x, y_1\rangle &= \langle (M^\perp) + x_0, y_1\rangle = \langle x_0, y_1\rangle = a_1 \\ \langle x, y_2\rangle &= \langle x_0, y_2\rangle = a_2 \\ &\vdots \\ \langle x, y_m\rangle &= \langle x_0, y_m\rangle = a_m \end{aligned}\right\}$$

Let's pretend these are constraints

Given constraints:

$$\begin{aligned} \langle x, y_1\rangle &= a_1 \\ \langle x, y_2\rangle &= a_2 \\ &\vdots \\ \langle x, y_m\rangle &= a_m \end{aligned}$$

what is the x of minimum norm that satisfies all those constraints?

Sounds oddly familiar to what we have done before

$x \in M$.

$$\boxed{x = c_1 y_1 + c_2 y_2 + \ldots + c_m y_m} \in M.$$

$$\begin{aligned} \langle x, y_1\rangle &= a_1 = \langle c_1 y_1 + c_2 y_2 + \ldots + c_m y_m, y_1\rangle \\ \langle x, y_2\rangle &= a_2 = \langle c_1 y_1 + c_2 y_2 + \ldots + c_m y_m, y_2\rangle \\ &\vdots \\ \langle x, y_m\rangle &= a_m = \langle c_1 y_1 + \ldots + c_m y_m, y_m\rangle \end{aligned}$$

$$\begin{bmatrix} \langle y_1, y_1\rangle & \langle y_2, y_1\rangle & \cdots & \langle y_m, y_1\rangle \\ \langle y_1, y_2\rangle & \langle y_2, y_2\rangle & \cdots & \langle y_m, y_2\rangle \\ & & \vdots & \\ \langle y_1, y_m\rangle & \cdots & & \langle y_m, y_m\rangle \end{bmatrix} \begin{bmatrix} c_1 \\ c_2 \\ \vdots \\ c_m \end{bmatrix} = \begin{bmatrix} a_1 \\ a_2 \\ \vdots \\ a_m \end{bmatrix}$$

$$\underbrace{\begin{bmatrix} \langle y_1, y_m \rangle & \cdots & \langle y_m, y_m \rangle \end{bmatrix}}_{} \begin{bmatrix} \langle m \rangle \\ \vdots \\ a_m \end{bmatrix}$$

$$\uparrow$$

$$\approx$$